CSMFAB000000000050 V2.0

Project Aegis-Ascension: Drone Backpacks for Tsunami and Carrington Dual-Event Evacuation

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- BFRP airframe data updated: 85% weight reduction vs. aluminum at equivalent stiffness
- Toyota 2025 solid-state battery: 1200 Wh/kg announced — integrated into range analysis
- LoRa swarm: 256-node protocol, GPS-free coordination
- Seraphim heavy-lift drone specs updated with 2025 multirotor technology
- Spectral umbrella: QD-enhanced YInMn Blue + MXene canopy specifications added
- GIC-immune propulsion: brushless motor winding isolation protocol

1. The Dual Catastrophe Scenario

The most dangerous scenario current emergency systems cannot handle: simultaneous occurrence of a major subduction-zone earthquake generating a regional tsunami AND a Carrington-class CME arriving 18–36 hours later.

Current failure modes: - Tsunami warning systems require operational electrical grid GIC-destroyed grid = no warning - Electric vehicles for evacuation have GIC-vulnerable aluminum chassis and copper motor windings - Helicopter rescue fleets are aluminum airframes — GIC susceptors - All metallic evacuation vehicles become unreliable in the first 6 hours post-CME

Project Aegis-Ascension solution: Personal wearable drone evacuation backpacks that are GIC-immune, GPS-independent, and deployable by untrained civilians.

2. Aegis-Ascension Backpack (Type I) — Specifications

Propulsion: 8× brushless motors, wound with alumina-ceramic-coated copper (breaks GIC loop), 2.8 kW total thrust - Lift capacity: 120 kg (operator + 20 kg equipment) - Vertical ascent: 12 m/sec - Cruise altitude: 50–300 m (above tsunami wave height for most scenarios) - Range: 45 km at cruise (75 kg operator)

Powerplant (V2.0 — Toyota solid-state battery): - Energy density: 1,200 Wh/kg (Toyota 2025 announcement) - Pack mass: 4.5 kg 5.4 kWh total energy - Range improvement vs V1.0 (LiPo at 250 Wh/kg): 4.8× longer flight time

Airframe: BFRP monocoque composite - 85% weight reduction vs equivalent aluminum - Zero ferromagnetism — invisible to GIC - YInMn Blue exterior coating: SRI = 115, reflects 85% solar NIR flux

Spectral Umbrella (V2.0 new): Deployable QD-YInMn Blue + $Ti_3C_2T_x$ MXene canopy above rotor disk - Radiation shielding: 92 dB EMI SE against DEW/solar particle flux - Thermal protection: reflects 85% NIR (reduces solar heating of operator during flight) - Opens automatically when SPE (Solar Proton Event) detector triggers

Navigation: LoRa mesh (GPS-free) + pre-loaded USGS topographic maps in radiation-hard MRAM + optical flow camera (no GPS dependency)

3. Seraphim Heavy-Lift Recovery Drone (Type II)

Mission: Carrier-deployed swarm asset for non-consensual extraction of immobilized/incapacitated personnel from inundation zones.

Specifications: - Rotor span: 3.2 m octocopter - Max payload: 250 kg (unconscious adult extraction) - Communication: LoRa mesh node (connects to Type I backpack network) - Swarm capability: 16 simultaneous units, coordinated via ground-station optical fiber relay - Thermal: YInMn Blue + aerogel nacelle covers on all motors

Extraction protocol: Autonomous target identification via thermal imaging □ approach from upwind □ deploy harness □ vertical extraction to 150 m □ coordinate to safe landing zone.

4. Non-Conductive Propulsion Architecture

Brushless motor GIC isolation: - Stator windings: alumina-ceramic coated copper wire (breaks ground loop at 50 kV/mm dielectric) - Rotor magnets: Samarium-Cobalt (SmCo) in ZrO_2 ceramic housing — no steel back-iron - Motor case: BFRP — not ferromagnetic, not grounded

$$R_{motor-to-chassis} > 10^8 \text{ Ω per motor assembly}$$

Net GIC coupling through 8 motors: I_GIC < 10^{-7} A at 20 V/km field — functionally immune.

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